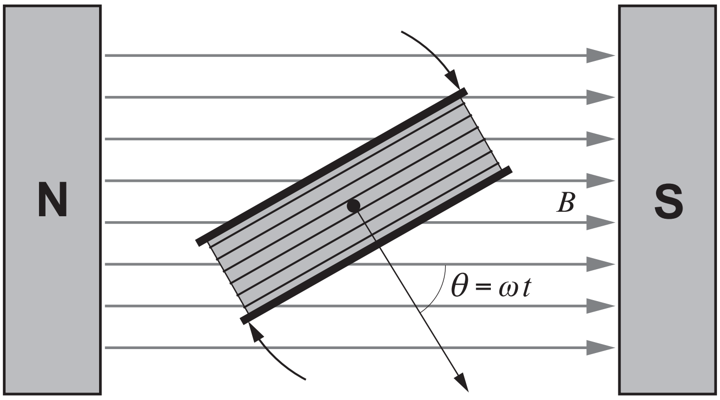
Examiner only

Option A – **Alternating Currents**

7. (a) A diagram of a rotating coil in a magnetic field is shown below.



The coil rotates 50 times per second in a field of flux density 0.215 T.
The coil has 2650 turns and cross-sectional area $2.35 \times 10^{-3}\,m^2$. (Note that $\theta = 0$ when $t = 0$)

(i) Calculate the peak induced emf in the coil. [2]

....

....

....

(ii) Calculate the flux linkage of the coil when $t = 5.0$ ms.
[Hint: put your calculator in radian mode.] [2]

....

....

....

(iii) Calculate the induced emf in the coil when $t = 5.0$ ms. [1]

....

....

Examiner only

(iv) Vanessa states that her answers to parts (ii) and (iii) are consistent with the coil having completed a quarter of a cycle. Determine whether or not she is correct. [3]

....

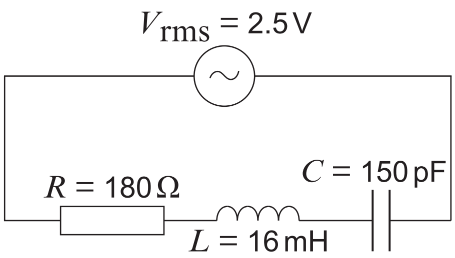
....

....

....

....

(b) (i) Calculate the resonance frequency of the following circuit. [2]



....

....

....

(ii) Explain why the rms current at resonance is approximately 14 mA. [2]

....

....

....

(iii) Calculate the rms current in the circuit when the frequency is 105 kHz. [4]

....

....

....

....

....

Examiner only

(iv) Explain why your answers to parts (b)(i), (b)(ii) and (b)(iii) suggest that the circuit has a high Q factor. [2]

....

....

....

(v) Thomas considers the effect, on this circuit, of increasing the capacitance only. He believes that this will decrease the Q factor. Determine whether or not he is correct. [2]

....

....

....

20